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|                 |                                                                                                                                                                                                                                                                                                                                 |
| <b>2(a)(i)</b>  | $T = \frac{2\pi r}{v}$ $= \frac{2\pi(1.75 \times 10^7)}{200}$ $= 5.50 \times 10^5 \text{ s}$ $= 6.36 \text{ days}$                                                                                                                                                                                                              |
| <b>2(a)(ii)</b> | <ol style="list-style-type: none"> <li>1. The relative position of Charon in the sky with respect to the space probe does not change so it is able to observe Charon at all times.</li> <li>2. The space probe would be able to observe the entire surface of Charon after one complete orbit of Charon about Pluto.</li> </ol> |

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| <b>2(b)</b>     | $g_p = \frac{GM}{r^2}$ $= \frac{(6.67 \times 10^{-11})(1.31 \times 10^{22})}{(1.20 \times 10^6)^2}$ $= 0.607 \text{ N kg}^{-1}$                                                 |
| <b>2(c)(i)</b>  | <p>Gain in KE = Loss in GPE</p> $\frac{1}{2}Mv^2 = 0 - \left( -\frac{GM_p M}{r} \right)$ $v^2 = \frac{2GM_p}{r}$ $v^2 = \frac{2GM_p r}{r^2}$ $v^2 = 2g_p r$ $v = \sqrt{2g_p r}$ |
| <b>2(c)(ii)</b> | $v = \sqrt{2g_p r}$ $= \sqrt{2(0.607)(1.20 \times 10^6)}$ $= 1210 \text{ m s}^{-1}$                                                                                             |
| <b>3(a)</b>     | A polarised wave is a transverse wave in which its oscillations are only in one                                                                                                 |